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NBA Cap and Trade Advisor

Cited, verified answers to salary cap and trade questions for the Toronto Raptors front office

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THE PROBLEM

Cap questions take hours of reading. The advisor answers in seconds, with sources.

676

pages in the 2023 NBA collective bargaining agreement

4

salary lines every move is checked against: cap, tax and two aprons

455

player contracts across 30 teams to keep track of

Today

An analyst searches the agreement and a spreadsheet by hand to answer “can we make this trade?”

With the advisor

Ask in plain English. Get the rule, the numbers for both teams, and a citation you can check. If it isn’t sure, it says so.

DATA AND RETRIEVAL

Two sources, handled differently

The CBA: searched

Why: the official rulebook, so every answer can cite it

- Split by article and section: 799 chunks
- Each chunk starts with its citation
- Searched locally, free and offline

Contracts: calculated

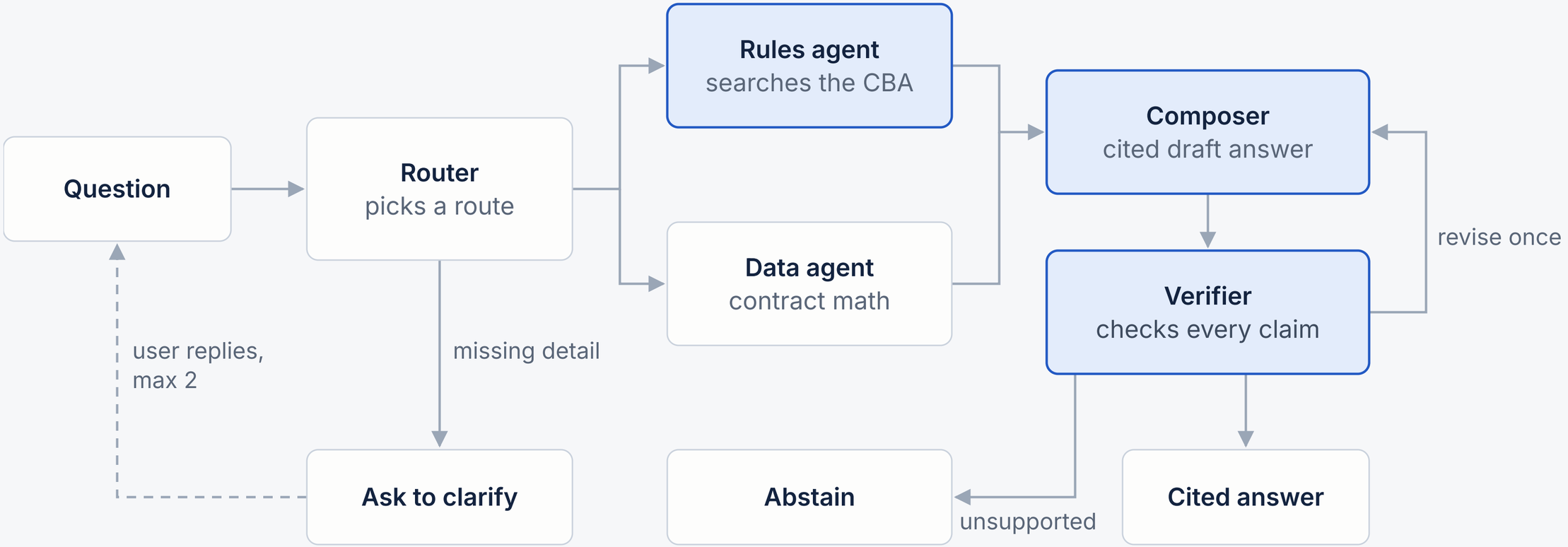
Why: public, league-wide, salaries through 2031-32

- 455 players across 30 teams
- 6 players listed twice, kept once
- 13 broken names fixed by hand

ARCHITECTURE

How a question flows through the system

Blue uses Gemini. White is plain code.



AGENTS AND SKILLS

Each agent has one job and its own tested tools

SEARCH + LLM

Rules agent

Finds the CBA sections

`search_cba`
`rewrite_query`

CODE ONLY

Data agent

Team numbers and trades

`get_team_payroll`
`get_apron_status`
`check_trade_side`

LLM

Composer

Writes the cited answer

`compose_answer`

CODE + LLM

Verifier

Checks every claim

`check_citations`
`check_numbers`
`check_with_model`

Router (code only): checks 5 rules in order and uses the first one that matches

1. The question says “him” but doesn’t name a player
→ **Ask who**

2. The question is a trade with players from 2 teams
→ **Both agents**

3. The question is a trade but doesn’t name the players
→ **Ask which players**

4. The question names a team and a numbers word, like “tax”
→ **Data agent**

5. The question is anything else
→ **Rules agent**

Agents never call each other. Each reads and writes one shared state.

STATIC VS DYNAMIC WORKFLOW

Dynamic runs only the steps a question needs

“Are we over the tax line?”

Static	Route	Search CBA	Numbers	Write	Verify	**Answers** · 4,699 tokens
Dynamic	Route	~~Search CBA~~	Numbers	Write	Verify	**Answers** · 1,093 tokens

“Can we trade him?”

Static	Route	Search CBA	Numbers	Write	Verify	**Refuses** · 1,421 tokens
Dynamic	Route	Ask: which player?				**Asks first** · 0 tokens

MODEL AND CONTEXT

Retrieval settings were measured, not guessed

Top-k: CBA chunks per question

Top-k	Correct	Retrieval	Tokens
3	23/25	5/6	40,038
5	24/25	5/6	50,916
8, chosen	**25/25**	**6/6**	**67,295**

Example: the free-agent trade rule ranks 6th. Top-5 never sees it, top-8 does.

Similarity threshold: 0.55

Question	Top score
Real rules questions	0.52 to 0.76
Off-topic (pizza)	0.19

Below 0.55: reword once, search again, refuse if still weak.

MODELS
Gemini 3.8 Flash writes and verifies. 3.5 Flash Lite rewords.

EVIDENCE
Only retrieved chunks and tool numbers reach the model, each labelled.

HISTORY
None kept. A clarifying reply is added to the question.

DESIGN CHOICES AND INFRASTRUCTURE

Each choice trades a little flexibility for answers you can check

Keyword router over an LLM classifier

Tested, free and instant. Misses phrasings it doesn't know.

Numbers from code over the model

Cap math is exact. Only answers what the tools cover.

Free checks before a paid check

Citations and numbers first. Checks the data match, not truth.

Refuse over guess

No confident wrong answers. Some good answers get refused.

Local search over a paid API

MiniLM and Chroma run free and offline. A smaller model.

Infrastructure

Runs on a laptop. Only Gemini is paid, about 2,400 tokens a question.

Main focus: clear testing. 216 unit tests check the code, and a 25-question eval checks the answers.

EVALUATION

Every answer is checked against one written first

25/25

correct, dynamic workflow, top-k 8

1. Write Questions and expected answers, before any run

2. Run One command asks them all and scores each one

3. Compare Results are saved to a file, so every change is measured

Question type	Result
Rules from the CBA	6/6
Team numbers	5/5
Rules + numbers (trades)	6/6
Vague, should ask	3/3
Missing info, should refuse	3/3
Clarify limit	2/2
Total	**25/25**

HALLUCINATION HANDLING

Four checkpoints stop unsupported claims before they reach the user

1

Vague question

Ask the user, at most twice, then refuse

2

Weak search

Reword the question once, then refuse

3

Strict prompt

Answer only from the evidence, never do math

4

Verifier

Reject fake citations or numbers, then double-check

Caught in testing

The model wrote **$19,500** instead of **$19,500,000**. The verifier’s number check caught it.

Every run has a limit

One retry, then refuse. Anything that breaks or runs too long also ends in a refusal.

LIVE DEMO

Six questions: what works, how it adapts, and where it fails

#	Question	Shows
1	How long after signing can a draft rookie be traded?	Cited answer
2	Can a team trade cash in a trade?	Cited answer
3	Can we trade him? (static, then dynamic)	Asks vs refuses
4	What is the Raptors payroll? (static, then dynamic)	Skipping work
5	When can a free agent be traded? (top-k 5, then 8)	Context change
6	Trade Quickley and Poeltl for Damian Lillard	A real failure

RECOMMENDATION

Pilot it now for rule lookups and trade checks, then close four gaps

Limitation today	Next step
Command line only	A simple web interface with answer, citations and steps
Contract data isn’t official, so Lillard lands on the wrong team	An official payroll source, and refuse trades for flagged players
Covers the 2026-27 season only	Future seasons: the data already runs through 2031-32
Tested on only 25 questions, each run once	Test more questions, run each several times, and average the results

DISCLOSURE

How it was built

AI TOOLS

I built this alongside Claude. It helped with explanations, code reviews, data cleaning, tests and slide drafts. **The design choices and trade offs were mine.**

SOURCES

- NBA and NBPA, 2023 Collective Bargaining Agreement
- Basketball Reference, player contracts, as of Nov 5, 2026
- NBA, official 2026-27 cap, tax and apron figures (pr.nba.com)